JET COURIER ? QA TESTING MANUAL

Pre-Release Verification & Offline-First Protocol Guide

Stack: Kotlin, Compose, Room SQLite, Firestore | Target: Google Play Production

1. SYSTEM ARCHITECTURE & OFFLINE-FIRST SYNCHRONIZATION

- Modern Native Architecture: 100% Kotlin Coroutines & Flow, declarative Jetpack Compose UI (Material 3), Room SQLite DB, Firebase Firestore, and Firebase Authentication.
- Single Source of Truth (SSOT): The local Room SQLite database serves as the absolute SSOT. All UI screens reactively observe Flow<List<T>> streams, ensuring zero wait-time on remote network calls.
- Instant Zero-Latency Execution: All actions (posting a flight, booking a package, sending a chat message, rating a peer) execute immediately on SQLite, providing a responsive 0ms user experience.
- Background Firestore Uplink: FirestoreSyncManager monitors events and asynchronously pushes changes to Google Cloud Firestore on Dispatchers.IO.
- Attribute-Based Access Control (ABAC): Private delivery chats are query-authorized via whereArrayContains('participantUids', currentUid), strictly mirroring Firestore security rules to protect user communications.
- Airport Network Resilience: Network failures, captive portals, and airplane mode trigger graceful fallbacks (withTimeout 15s) with zero UI freezes or frozen progress bars.

2. DATA MODELS & DATABASE SCHEMA REFERENCE

Table: flights (Firestore Collection: flights)

Field	SQLite Type	Kotlin / Firestore	Constraints & Notes
id	INTEGER	Int	Primary Key, Auto-Generate
flightNumber	TEXT	String	e.g. BA-177, SQ-308, LH-430
fromAirportCode	TEXT	String	3-letter IATA code (e.g. LHR, FRA)
toAirportCode	TEXT	String	3-letter IATA code (e.g. JFK, BUD)
departureDate	TEXT	String	ISO format: YYYY-MM-DD
capacityKg	REAL	Double	Max luggage capacity in kg
contactPhone	TEXT	String	Traveler contact phone with code
contactEmail	TEXT	String	Verified contact email
notes	TEXT	String	Meeting location, baggage notes
userRating	REAL	Double	Community rating (0.0 to 5.0)
ratingCount	INTEGER	Int	Total review count
isMyOffer	INTEGER	Boolean	true if created by local user
userId (cloud)	?	String	Firebase Auth UID of flyer

Table: packages (Firestore Collection: packages)

Field	SQLite Type	Kotlin / Firestore	Constraints & Notes
id	INTEGER	Int	Primary Key, Auto-Generate
title	TEXT	String	Shipment title headline
fromCityOrAirport	TEXT	String	Origin city or IATA airport
toCityOrAirport	TEXT	String	Destination city or airport
weightKg	REAL	Double	Parcel weight in kilograms
dimensions	TEXT	String	e.g. 20x15x10 cm, A4 Envelope
rewardUsd	REAL	Double	Monetary compensation ($ USD)
deadline	TEXT	String	Delivery deadline (YYYY-MM-DD)
urgency	TEXT	String	LOW, NORMAL, HIGH, URGENT
senderName	TEXT	String	Shipper full name
status	TEXT	String	OPEN, MATCHED, IN_TRANSIT, DELIVERED
matchedFlightId	INTEGER	Int?	Foreign key referencing flights.id
isMyOffer	INTEGER	Boolean	true if authored by local user
createdAt	INTEGER	Long	Millisecond Unix timestamp
userId (cloud)	?	String	Firebase Auth UID of shipper

Table: delivery_reviews (Firestore Collection: delivery_reviews)

Field	SQLite Type	Kotlin / Firestore	Constraints & Notes
id	INTEGER	Int	Primary Key, Auto-Generate
packageId	INTEGER	Int	Indexed. Linked package ID
flightId	INTEGER	Int?	Indexed. Linked flight ID
reviewerRole	TEXT	String	SENDER or FLYER
reviewerName	TEXT	String	Display name of reviewer
targetName	TEXT	String	Name of reviewed user
rating	INTEGER	Int	Star rating: 1 to 5 integer
feedbackComment	TEXT	String	Review message text
selectedTags	TEXT	String	e.g. On-Time Delivery, Safe Handoff
createdAt	INTEGER	Long	Millisecond Unix timestamp

Table: chat_messages (Firestore Collection: chat_messages)

Field	SQLite Type	Kotlin / Firestore	Constraints & Notes
id	INTEGER	Int	Primary Key, Auto-Generate
packageId	INTEGER	Int	Indexed. Deal chat conversation ID
senderRole	TEXT	String	SENDER, FLYER, or SYSTEM
senderName	TEXT	String	Display name of sender
isFromSenderRole	INTEGER	Boolean	true if message from shipper
text	TEXT	String	Message body content
timestamp	INTEGER	Long	Millisecond Unix timestamp
messageType	TEXT	String	TEXT, MEETUP, WEIGHT, STATUS, etc.
participantUids	?	List<String>	Cloud array for ABAC authorization

3. APPLICATION STATES, ENUM VALUES & ROLES

- Package Status (OPEN): Initial state. Package is listed publicly seeking couriers; matchedFlightId is null.
- Package Status (MATCHED): Courier accepts parcel or shipper assigns courier. Chat channel unlocks between peers.
- Package Status (IN_TRANSIT): Luggage handover confirmed at terminal; courier boards flight with parcel.
- Package Status (DELIVERED): Parcel arrives at destination; recipient confirms receipt; rating prompt triggered.
- Chat Message Types: TEXT (standard chat), MEETUP_PROPOSAL (terminal meetup coordination), WEIGHT_CONFIRMATION (scale weight verification), STATUS_ALERT (flight state updates), SECURITY_VERIFICATION (IATA compliance), PHOTO_RECEIPT (baggage handover receipt).
- User Roles: SENDER (parcel owner/shipper), FLYER (traveling passenger courier), SYSTEM (automated trust verification and status notifications).

4. KEY MANUAL TEST SCENARIOS (STEP-BY-STEP QA)

Scenario 1: Authentication & Account Recovery

1. Launch app on clean install -> Select 'Create Account' tab.
2. Enter email, password (min 6 chars), confirm password -> Tap 'Create Account'. Confirm successful dashboard entrance.
3. Sign out -> Tap 'Forgot Password?' -> Enter email -> Tap 'Send Reset Link'. Confirm green 'Email Sent!' dialog.
4. Tap 'Continue with Google' -> Select account in Credential Manager bottom sheet. Confirm silent instant sign-in.

Scenario 2: Creating a Flight & Package

1. Open 'My Flights' -> Tap '+' (FAB) -> Enter Flight BA-249, LHR -> GIG, capacity 5.0 kg -> Publish. Confirm flight appears in feed with 'My Listing' badge.
2. Open 'Send Package' -> Enter Title 'Camera Lens', LHR -> GIG, 1.5 kg, $180 reward -> Post. Confirm package appears with 'OPEN' status.

Scenario 3: Courier Matching & Real-Time Chat

1. Open package 'Camera Lens' -> Tap 'Accept Delivery / Assign Flight' -> Choose flight BA-249. Confirm status changes to 'MATCHED'.
2. Tap 'Open Delivery Chat' -> Send 'Meeting at Gate B32' -> Tap quick proposal 'Propose Airport Meetup'. Confirm messages stream to both peers in real time without permission errors.

Scenario 4: Airport Offline Mode Stress-Test (Crucial QA Check)

1. Put Android test device into Airplane Mode (Wi-Fi: OFF, Cellular: OFF).
2. In active delivery chat, send messages, change package status to 'IN_TRANSIT', and submit a 5-star courier rating.
3. Verify: Zero progress bar hangs, zero ANR dialogs. All updates save instantly to local Room SQLite.
4. Restart app while still in Airplane Mode: verify package status remains 'IN_TRANSIT' and chat messages persist.
5. Turn OFF Airplane Mode (reconnect to LTE/Wi-Fi): observe Logcat tag 'FirestoreSyncManager'. Confirm cloud sync updates document to 'IN_TRANSIT' and pending review uploads automatically.

5. KNOWN CONSTRAINTS & BORDER CASES FOR QA

- Time Zone Offsets: Couriers cross time zones (e.g., Tokyo UTC+9 to NY UTC-5). Departure dates (YYYY-MM-DD) must remain fixed to flight origin airport schedule regardless of local phone time zone changes.
- Network Drops & Timeouts: Airport Wi-Fi networks frequently drop. All repository calls enforce withTimeout(15_000L) and catch SocketTimeoutException, preventing hanging spinners.
- Weight Limit Enforcement: A courier flight with capacityKg = 3.0 kg must reject package assignments weighing > 3.0 kg, displaying a validation warning.
- Database Migrations: Room schema is locked at Version 5 without destructive fallback. Upgrading APKs from older builds must never drop tables or wipe user data.